

PUBLICLY
AVAILABLE
SPECIFICATION

IEC
PAS 61249-3-1

Pre-Standard

First edition
2007-05

**Materials for printed boards and other
interconnecting structures –**

**Part 3-1:
Copper-clad laminates for flexible boards
(adhesive and non-adhesive types)**



JPCA

Reference number
IEC/PAS 61249-3-1:2007(E)



THIS PUBLICATION IS COPYRIGHT PROTECTED

Copyright © 2007 IEC, Geneva, Switzerland

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from either IEC or IEC's member National Committee in the country of the requester.

If you have any questions about IEC copyright or have an enquiry about obtaining additional rights to this publication, please contact the address below or your local IEC member National Committee for further information.

IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland
Email: inmail@iec.ch
Web: www.iec.ch

About the IEC

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

About IEC publications

The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigenda or an amendment might have been published.

- Catalogue of IEC publications: www.iec.ch/searchpub

The IEC on-line Catalogue enables you to search by a variety of criteria (reference number, text, technical committee,...). It also gives information on projects, withdrawn and replaced publications.

- IEC Just Published: www.iec.ch/online_news/justpub

Stay up to date on all new IEC publications. Just Published details twice a month all new publications released. Available on-line and also by email.

- Customer Service Centre: www.iec.ch/webstore/custserv

If you wish to give us your feedback on this publication or need further assistance, please visit the Customer Service Centre FAQ or contact us:

Email: csc@iec.ch
Tel.: +41 22 919 02 11
Fax: +41 22 919 03 00

PUBLICLY
AVAILABLE
SPECIFICATION

IEC
PAS 61249-3-1

Pre-Standard

First edition
2007-05

**Materials for printed boards and other
interconnecting structures –**

**Part 3-1:
Copper-clad laminates for flexible boards
(adhesive and non-adhesive types)**

LICENSED TO MECON Limited - RANCHI/BANGALORE
FOR INTERNAL USE AT THIS LOCATION ONLY, SUPPLIED BY BOOK SUPPLY BUREAU.



Commission Electrotechnique Internationale
International Electrotechnical Commission
Международная Электротехническая Комиссия

PRICE CODE

V

For price, see current catalogue

CONTENTS

FOREWORD	4
1 Scope	6
2 Normative references	6
3 Terms and definitions	6
4 Designation of copper-clad laminates.....	6
4.1 Copper-clad laminate.....	7
4.2 Base materials.....	7
4.3 Thickness of the base material.....	7
4.4 Types of adhesives	8
4.5 Thickness of base materials and adhesives.....	8
4.6 Type of copper foil	9
4.7 Grade of copper foil	9
4.8 Copper-foil thickness	9
4.9 Types of profiles	10
4.10 Surface treatment to increase copper adhesivity and anti-rust	10
4.11 Symbol for flammability.....	10
5 Observation.....	10
5.1 Base film	10
5.2 Copper foil	11
5.3 CCL	11
6 Size	12
6.1 Base film	12
6.1.1 Thickness and its allowance.....	12
6.2 Copper foil	12
6.2.1 Thickness and its allowance.....	12
6.3 Adhesives	13
6.3.1 Adhesives.....	13
6.4 Copper-clad laminates.....	13
6.4.1 Thickness and its allowance.....	13
6.4.2 Sheet dimension and its allowance.....	13
6.4.3 Role dimension and its allowance.....	13
7 Properties	13
7.1 Base film	13
7.2 Copper foil	15
7.3 CCL	17
8 Package and labelling.....	25
Annex A (normative) Roughness test.....	26
Annex B (normative) Dimensional stability test	27
Appendix.....	30
Figure B.1 – Test pattern for the dimensional stability test.....	27
Table 1 – Copper-clad laminates.....	7
Table 2 – Base materials	7
Table 3 – Thickness of base material	8
Table 4 – Adhesives	8

Table 5 – Thickness of adhesives	9
Table 6 – Grade of copper foil.....	9
Table 7 – Thickness of copper foil (Types E and R)	10
Table 8 – Types of profiles	10
Table 9 – Appearance of CCL	11
Table 10 – Thickness and its allowance of base film	12
Table 11 – Thickness and its allowance of copper foil (Types E and R)	12
Table 12 – Thickness and its allowance of adhesives	13
Table 13 – Properties of polyimide film.....	14
Table 14 – Properties of copper foil (type E)	15
Table 15 – Properties of copper foil (type R)	16
Table 16 – Properties of CCL Adhesive type (three layers)/polyimide film base	17
Table 17 – Properties of CCL Non-adhesive type (two layers)/polyimide film base + copper foil (casting)	19
Table 18 – Properties of CCL Non-adhesive type (two layers)/polyimide film base + copper foil (sputter/plating)	21
Table 19 – Properties of CCL Non-adhesive type (two layers)/polyimide film base + copper foil (laminate)	23

INTERNATIONAL ELECTROTECHNICAL COMMISSION

MATERIALS FOR PRINTED BOARDS AND OTHER INTERCONNECTING STRUCTURES –

Part 3-1: Copper-clad laminates for flexible boards (Adhesive and non-adhesive types)

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC provides no marking procedure to indicate its approval and cannot be rendered responsible for any equipment declared to be in conformity with an IEC Publication.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

A PAS is a technical specification not fulfilling the requirements for a standard but made available to the public.

IEC-PAS 61249-3-1 was submitted by the JPCA (Japan Electronics Packaging and Circuits Association) and has been processed by IEC technical committee 91: Electronics assembly technology.

The text of this PAS is based on the following document:

This PAS was approved for publication by the P-members of the committee concerned as indicated in the following document:

Draft PAS	Report on voting
91/616/NP	91/644/RVN

Following publication of this PAS, which is a pre-standard publication, the technical committee or subcommittee concerned will transform it into an International Standard.

This PAS shall remain valid for an initial maximum period of three years starting from 2007-05. The validity may be extended for a single three-year period, following which it shall be revised to become another type of normative document or shall be withdrawn.

MATERIALS FOR PRINTED BOARDS AND OTHER INTERCONNECTING STRUCTURES –

Part 3-1: Copper-clad laminates for flexible boards (Adhesive and non-adhesive types)

1 Scope

This PAS specifies the properties of copper-clad laminates used for flexible boards for both adhesive and non-adhesive types.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

JPCA-TD01, *Terms and definition for printed circuits*

JIS C 5603, *Terms and definition for printed circuits*

JIS C 6471, *Test methods of copper-clad laminates for flexible printed wiring boards*

JIS C 6472, *Copper-clad laminates for flexible printed wiring boards (Polymer film, Polyimide film)*

JIS C 6515, *Copper foil for printed wiring boards*

IEC 60194, *Printed board design, manufacture and assembly – Terms and definitions*

IPC-4204, *Flexible Metal-Clad Dielectrics for Use in Fabrication of Flexible Printed Circuitry*

ASTM D149, *Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies*

ASTM D150, *Standard Test Methods for AC Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulation*

ASTM D882, *Standard Test Method for Tensile Properties of Thin Plastic Sheeting*

3 Terms and definitions

For the purposes of this document, the following terms and definitions, as well as those mentioned in IEC 60194, JIS C 5603 and JPCA-TD01, apply.

3.1

machine direction (MD)

longitudinal direction in production of film, copper foil, and copper-clad laminate

3.2

transverse direction (TD)

transverse direction in production of film, copper foil, and copper-clad laminate

4 Designation of copper-clad laminates

The designation of types of laminates shall be made in the following way. Constituent designations are connected by hyphens.

Symbol for identifying copper-clad laminate	Symbol for base material and its thickness	Symbol for adhesive and its thickness	Symbol for type and grade of copper-foil and thickness	Symbol for profile and treatment	Symbol for flammability
(3.1)	(3.2) (3.3)	(3.4) (3.5)	(3.6) (3.7) (3.8)	(3.9) (3.10)	(3.11)
Example: ADF	- E6	- TA	- E1/12/12	- V1	- F

4.1 Copper-clad laminate

The symbol to show the types of flexible printed wiring boards (hereinafter called CCL) shall be as given in Table 1.

Table 1 – Copper-clad laminates

Symbol	Copper-clad laminate
ADF	Adhesive type (3-layer) CCL
ALF	Non-adhesive type (2-layer) CCL

4.2 Base materials

The symbol to designate the base material shall be as given in Table 2.

Table 2 – Base materials

Symbol	Base material
E	Polyimide
B	Polyethylene terephthalate (PET)
A	Polyvinyl fluoride (PVF)
C	Ethylene fluoride-propylene copolymer (FEP)
D	Polytetra fluoroethylene (PTFE)
F	Aramid
G	Polyamide-imide
H	Epoxy
J	Polyether imide
K	Polysulfone
L	Polyethylene naphthalate (PEN)
M	Liquid crystal polymer (LCP)
X	Others

4.3 Thickness of the base material

The symbol to designate the thickness of base material shall be as given in Table 3.

Table 3 – Thickness of base material

Symbol	Thickness of base material µm
0	No base material
1	25
2	50
3	75
4	100
5	125
6	12,5
7	38
8	17,5
9	10
10	7,5
11	20
12	40
X	Other thickness
NOTE A thickness of 15 µm is an example of another thickness.	

4.4 Types of adhesives

The symbol to designate the adhesive shall be as given in Table 4.

Table 4 – Adhesives

Symbol	Adhesive
L	Epoxy resin
M	Acrylate
N	Polyester
O	No adhesive
P	Butyl phenol
R	Ethylene fluoride–propylene copolymer (FEP)
S	Nitrile phenol
T	Polyimide
U	Polyether
W	Perfluoroalkoxy (PFA)
Y	Cyanate ester
X	Other thickness

4.5 Thickness of base materials and adhesives

The symbol to designate the thickness of base material and adhesives shall be as given in Table 5.

Table 5 – Thickness of adhesives

Symbol	Thickness of adhesive μm
O	No adhesive
A	2,5
B	5
C	7,5
D	10
E	13
F	15
H	18
J	20
K	23
L	40
M	25
X	Other thickness

4.6 Type of copper foil

The symbol to designate the type of copper foil shall be as follows.

Type E: Electrodeposited copper foil

Type R: Rolled copper foil

Type O: Other type

4.7 Grade of copper foil

The symbol to designate the grade of copper foil shall be as given in Table 6. A slash shall be used between the symbols of grade and the thickness of the copper foil.

Table 6 – Grade of copper foil

Symbol	Type of copper foil
E1	Standard electrodeposited copper foil
E2	High-ductility electrodeposited copper foil
E3	High temperature elongation electrodeposited copper foil
R1	As-rolled wrought copper foil
R2	Light cold-rolled wrought copper foil
R3	Annealed wrought copper foil
O	Copper foil made by means other than rolling or electrolysis

4.8 Copper-foil thickness

The symbol to designate the thickness of copper foil shall be as given in Table 7. The thicknesses of the copper foils on both surfaces of a base (double-sided laminate) are expressed by two numbers separated by a slash. The thickness of copper foil of single-sided laminate is expressed with "0" after the slash. In the case of double-sided CCL with different copper foil thickness on both sides of the laminate, the thicker copper foil thickness shall be stated first in front of the slash. The designation for type O is to be agreed upon between the supplier and the user.

Table 7 – Thickness of copper foil (Types E and R)

Symbol	Nominal thickness μm	Mass designation g/m^2
Less than 5	Less than 5	To be agreed upon between the supplier and the user
9	9	76
12	12	107
18	18	152
25	25	230
35	35	305
50	50	445
70	70	610

4.9 Types of profiles

The symbol to designate the profile of copper foil to increase the adhesivity of the copper surface shall be as given in Table 8.

Table 8 – Types of profiles

Designation	Type	Ten-point height of roughness profile (Rz) μm
S	Standard	Less than 14*
L	Flatter than S (low profile)	Less than 10
V	Flatter than L (very low profile)	Less than 5
Z	Flatter than Z (ultra low profile)	Less than 2,5
* Applicable range: Thickness of less than 70 μm		
NOTE The measurement of the surface roughness is made by a stylus instrument. Use of an optical laser surface measurement is based on agreement between the user and the supplier.		

4.10 Surface treatment to increase copper adhesivity and anti-rust

The surface treatment for the purpose of increasing copper adhesivity and for anti-rusting may be applied on either both sides or a single side of the CCL. The symbol for the surface treatment shall be as follows.

0. No surface treatment for adhesivity increase, but both surfaces treated for anti-rusting.
1. Surface treatment for adhesivity increase on one surface, and treated for anti-rusting on both sides.
2. Surface treatment for adhesivity increase on both surfaces, and also treated for anti-rusting on both sides.

4.11 Symbol for flammability

An English capital letter “F” shall be designated when CCL complies the requirement of flammability stated in 6.3 (properties of CCL).

5 Observation

5.1 Base film

There shall be no break, hole, or wrinkle by folding in base film. There shall not be any defect that is harmful for the performance of CCL such as scratch, dent, surface damage, or any contamination.

5.2 Copper foil

The surface appearance shall comply with JIS C 6515, Clause 9. The surface condition for type O shall be agreed upon between the user and the supplier.

5.3 CCL

The appearance of the CCL shall comply with Table 9.

Table 9 – Appearance of CCL

Appearance	General purpose		High-quality products	
Copper-foil surface	(1) No swell nor wrinkle			
	(2) Corrosion product, discoloration, or adhered foreign substance on the surface shall be easily removed using diluted chloric acid of 1 mol/dm ³ , or other appropriate solvent			
	(3) There shall be no harmful scratches that may affect the performance of CCL			
	(4) The number of pinholes with a diameter larger than 0,10 mm but smaller than 0,25 mm in the area specified in JIS C 6471, 6.1.2, shall be less than one. There shall be no pin hole with a diameter larger than 0,25 mm. The number of pinholes with a diameter smaller than 0,10 mm shall be agreed between user and supplier		(4) The number of pinholes with a diameter larger than 0,10 mm but smaller than 0,15 mm in the area specified in JIS C 6471, 6.1.2, shall be less than one. There shall be no pin hole with a diameter larger than 0,15mm. The number of pinholes with a diameter smaller than 0,10mm shall be agreed between user and supplier	
	(5) The dents in the area specified in JIS C 6471, 6.1.2, are evaluated by the total points for dents according to the following table. The total points for dents shall be no greater than 35. The dents smaller than 0,25 mm shall be agreed between user and supplier		(5) The dents in the area specified in JIS C 6471, 6.1.2, are evaluated by the total points for dents according to the following table. The total points for dents shall be no greater than 25. There shall be no dent with a diameter larger than 1,00 mm. The dents smaller than 0,10 mm shall be agreed between user and supplier	
	Maximum diameter of dent and points		Maximum diameter of dent and points	
	Maximum diameter of dent mm	Point	Maximum diameter of dent mm	Point
	0,25 ≤ <0,50	2	0,10 ≤ <0,25	1
	0,50 ≤ <0,75	4	0,25 ≤ <0,50	2
	0,75 ≤ <1,00	7	0,50 ≤ <0,75	4
1,00 ≤	30	0,75 ≤ <1,00	20	
Base film surface	The surface shall be flat, and there shall be no swell, crack, or foreign substance that affects the performance of the material, dirt, dust, colour non-uniformity, cut, unevenness, or striped pattern			
Copper foil removed surface	There shall be no harmful cut, void, colour non-uniformity, nor striped pattern. Copper particles, foreign substances, and dust shall easily be removed by diluted hydrochloric acid of 1 mol/dm ³ or other appropriate solvent			

6 Size

6.1 Base film

6.1.1 Thickness and its allowance

The thickness and its allowance of base film shall comply with Table 10.

Table 10 – Thickness and its allowance of base film

Nominal thickness μm	Allowance %
7,5	To be agreed between user and supplier
10	
12,5	
17,5	± 16
20	
25	± 15
38	
40	
50	
75	± 12
100	
125	

6.2 Copper foil

6.2.1 Thickness and its allowance

The thickness and its allowance of copper foil shall comply with Table 11. The thickness and its allowance for copper foil of type shall be agreed between user and supplier.

Table 11 – Thickness and its allowance of copper foil (Types E and R)

Nominal thickness μm	Weight g/m^2	Specification	
		Allowance of thickness and weight %	
		Class I	Class II (for flexible boards)
Less than 5	To be agreed between user and supplier	To be agreed between user and supplier	To be agreed between user and supplier
9	76	± 10	± 5
12	107		
18	152		
25	230		± 10
35	305		
50	445		
70	610		

The measurement of copper foil with a thickness of less than 5 μm shall be made by the weight method (IPC-TM-650-2.2.12.2).

6.3 Adhesives

6.3.1 Adhesives

The thickness and its allowance of copper foil shall comply with Table 12.

Table 12 – Thickness and its allowance of adhesives

Thickness μm	Allowance %
<20	± 15
$20 \leq$	± 10

6.4 Copper-clad laminates

6.4.1 Thickness and its allowance

- The thickness of CCL with adhesive layer (three layers) shall be expressed by the sum of the thickness of the base film to be used, that of the copper foil, and that of the adhesive. The allowance for the thickness of CCL shall be within ± 20 % of the total thickness.
- The thickness of CCL without adhesive layer (two layers) shall be expressed by the sum of the thickness of the base film to be used, that of the copper foil, and that of the adhesive. The allowance for the thickness of CCL shall be within ± 20 % of the total thickness.

6.4.2 Sheet dimension and its allowance

The dimension of CCL in sheet form (length $\times$ width) shall be agreed upon between the user and the supplier.

6.4.3 Role dimension and its allowance

The dimension of CCL in role form (length $\times$ width) shall be agreed upon between the user and the supplier.

7 Properties

7.1 Base film

The properties of polyimide film shall be as specified in Table 13. The properties of other base film shall be agreed upon between the user and the supplier.

Table 13 – Properties of polyimide film

Item		Unit	Nominal thickness μm											Test method	
			7,5	10	12,5	17,5	20	25	38	40	50	75	100		125
Resistivity		Ωcm	To be agreed between user and supplier	10 ¹⁵ ≤											JIS C 2318 or ASTM D149
Breakdown voltage	Average	kV		2,5 ≤			4 ≤			5 ≤	7 ≤	9 ≤	11 ≤	JIS C 2318 or ASTM D149	
	Minimum			—			—			2 ≤	3 ≤	4 ≤	5 ≤		
Permittivity	1MHz	—		≤ 4,0											IPC-TM-650 2.5.5.3 or ASTM D150
Loss tangent	1MHz	—		≤ 0,05											
Tensile strength		N/ mm ²		137 ≤			146 ≤								JIS C 2318 or ASTM D882
Elongation		%		25 ≤											
Shrinkage by heating	200°C	%		≤ 0,30											IPC-TM-650 2.2.4 or ASTM D882
Flatness		—		To be agreed between user and supplier											IEC 674-2

7.2 Copper foil

Properties of copper foil shall be as specified in Tables 14 and 15. The properties of copper foil of type O shall be agreed upon between the user and the supplier.

Table 14 – Properties of copper foil (type E)

Item	Unit	Thickness Nominal μm	Specification						Test method JIS C 6515
			Room temperature			High temperature			
			23 °C			180 °C			
Purity	%		99,8 ≤						5. According to the test method (copper purity) for “untreated copper foil, type E”
Surface roughness	μm		≤ 0,43						Surface roughness shall be expressed by the averaged roughness at the centreline (Ra) in accordance with Annex A of this standard
Resistance (maximum)	mΩ	5 9 12 18 25 35 50 70	25,0 15,0 10,4 7,1 4,7 2,5 2,4 1,7						6. According to the test method (electrical) for electrical resistance for the maximum test specimen
Tensile strength (minimum)	N/mm ²		E1	E2	E3	E1	E2	E3	7. According to the test method for tensile strength
		5	—	—	—	—		—	
		9	105	105	105			130	
		12	105	105	105			130	
		18	105	105	105			130	
		25	150	150	150			135	
		35	210	210	210			140	
		50	210	210	210			155	
		70	210	210	210			175	
Elongation (minimum)	%		E1	E2	E3	E1	E2	E3	7. According to the test method for tensile strength
		5	—	—	—	—		—	
		9	2	5	2			2	
		12	2	5	2			2	
		18	2	5	2			2,5	
		25	2,5	7,5	2,5			3	
		35	3	10	3			3	
		50	3	12,5	3			3	
		70	3	15	3			3	
Resistivity to bending	Cycle	5	—						According to JIS P 8115 (MIT test method) Curvature radius: 2 mm Load: 500 g . 175 cycles/min Room temperature and after anneal of 1 h at 180 °C
		9 12 18	300 ≤						
		25 35	100 ≤						
		50 70	10 ≤						

Table 15 – Properties of copper foil (type R)

Item	Unit	Thickness Nominal μm	Specification				Test method JIS C 6515
			Room temperature 23 °C		High temperature 180 °C		
Purity	%		99,9 ≤				5. According to the test method for (copper purity), “untreated copper foil, type E”
Surface roughness	μm		≤ 0,25				Surface roughness shall be expressed by the averaged roughness at the centreline (Ra) in accordance with Annex A of this standard
Resistance (maximum)	mΩ	5 9 12 18 25 35 50 70	22,5 13,4 9,6 6,7 4,5 3,4 2,3 1,7				6. According to the test method (electrical) for the electrical resistance for the maximum test specimen
Tensile strength (minimum)	N/mm ²		R1	R2	R3	R1 R2 R3	7. According to the test method for tensile strength
		5	—	—	—	—	
		9	350	—	105		
		12	350	—	105		
		18	350	—	105		
		25	350	—	120		
		35	350	180	140		
		50	350	180	155		
70	350	180	155				
Elongation (minimum)	%		R1	R2	R3	R1 R2 R3	7. According to the test method for tensile strength
		5	—	—	—	—	
		9	0,5	—	5		
		12	0,5	—	5		
		18	0,5	—	5		
		25	0,5	—	7,5		
		35	0,5	10	10		
		50	0,5	12,5	15		
70	1,0	15	20				
Resistivity to bending	Cycle	5	—				According to JIS P 8115 (MIT test method) Curvature radius: 2 mm, Load: 500g , 175 cycles/min Room temperature and after anneal of 1 h at 180 °C
		9	600 ≤				
		12					
		18					
		25	200 ≤				
		35					
		50	20 ≤				
		70					

7.3 CCL

The properties of CCL shall be as specified in Tables 16 and 17. The properties of other CCL including high Tg and halogen-free materials shall be agreed upon between the user and the supplier.

**Table 16 – Properties of CCL –
Adhesive type (three layers)/polyimide film base**

Item				Unit	Specification		Test method JIS C 6471
Observation				—	According to Table 9		According to 6.1 (observation)
Size	Thickness			—	According to 5.4.1		According to 6.2.1 (thickness)
	Width				According to 5.4.2 and 5.4.3		According to 6.2.2 (width)
	Length						According to 6.2.3 (length)
Resistivity	Normal			Ω · cm	10 ¹³ ≤		According to 7.1 (resistivity)
	After water vapour adsorption				10 ¹² ≤		
Surface resistivity	Normal			Ω	10 ¹¹ ≤		According to 7.2 (surface resistivity)
	After water vapour adsorption				10 ¹⁰ ≤		
Surface withstanding voltage AC 500 V, 1 min				—	There shall be no flash-over		According to 7.3 (surface withstanding voltage)
Interlayer withstanding voltage AC 500 V, 1 min				—	There shall be no breakdown		According to 7.4 (interlayer withstanding voltage)
Dielectric constant (1 MHz)				—	≤ 4,0		According to 7.5 (dielectric constant and loss tangent)
Loss tangent (1 MHz)				—	≤ 0,07		
Peel strength of copper foil (Thickness of polyimide film: 25 μm)	Thickness of copper foil			—	8 ≤	< 35	According to 8.1 (peel strength of copper foil) (when the thickness of the copper foil is less than 8 μm, electroplate copper on the foil as to attain the thickness of more than 8 μm)
	Item	Normal		N/mm	0,7 ≤	0,8 ≤	
		After heating			0,6 ≤		
		Dipped into solder			—		
		Dipped into chemicals			0,5 ≤	0,6 ≤	
Bending (Bending radius is R = 2, 0 mm for thickness of polyimide film: 25 μm)	Copper foil	Type E	Copper foil	18	Cycle	300 ≤	According to 8.2 (bending)
				35		250 ≤	
				18		500 ≤	
				35		400 ≤	

Table 16 – Properties of CCL (continued)

Item			Unit	Specification		Test method JIS C 6471
Flammability			—	Specimen shall satisfy the following specification		According to 9.1 (flammability)
				Item	Specification	
				(1) Flaming time	Within 10 s for each time, and total of less than 50 s for 10 trials	
				(2) Flaming and glowing times	Less than 30 s for the total of second trials	
				(3) Flaming or glowing to the clamp, or marking line	There shall be no flaming or glowing	
				(4) Drop from specimen that ignite cotton.	There shall be no ignition	
				NOTE Repeat the test when one specimen does not satisfy the test items (1) to (4), and the total flaming time of ten trials is between 51s and 55 s. The specimens shall satisfy the specification in the repeated test		
Resistance to chemicals			—	There shall be no swell nor delamination		According to 9.2 (resistance to chemicals)
Resistance to soldering			—	There shall be no considerable shrinkage, and no delamination of copper foil and base, nor swell		According to 9.3 (resistance to soldering heat)
Solderability			—	The area of copper foil with good solderability shall be over 95 % of the foil.		According to 9.4 (solderability)
Chemicals for processing of copper foil			—	The chemical for copper-foil treatment shall not affect etching of CCL, soldering nor adhesiveness of electrolytic plating		According to 9.5 (chemicals for processing of copper foil)
Dimensional stability (Thickness of polyimide film: 25 μm)	After etching and drying	MD	%	± 0,15		According to 9.6 (dimensional stability)
		TD				
	After etching and heating	MD		± 0,20		
		TD				
	After heating	MD		±0,10		According to Method D in Annex B of this standard
		TD				

**Table 17 – Properties of CCL –
Non-adhesive type (two layers)/polyimide film base + copper foil (casting)**

Item				Unit	Specification			Test method JIS C 6471		
Observation				—	According to Table 9			According to 6.1 (observation)		
Size	Thickness			—	According to 5.4.1			According to 6.2.1 (thickness)		
	Width				According to 5.4.2 and 5.4.3			According to 6.2.2 (width)		
	Length							According to 6.2.3 (length)		
Resistivity	Normal			$\Omega \cdot \text{cm}$	$10^{12} \leq$			According to 7.1 (resistivity)		
	After water vapour adsorption				$10^{11} \leq$					
Surface resistivity	Normal			Ω	$10^{11} \leq$			According to 7.2 (surface resistivity)		
	After water vapour adsorption				$10^{10} \leq$					
Surface withstanding voltage AC 500 V, 1 min				—	There shall be no flash-over			According to 7.3 (surface withstanding voltage)		
Interlayer withstanding voltage AC 500 V, 1 min				—	There shall be no breakdown			According to 7.4 (interlayer withstanding voltage)		
Dielectric constant (1 MHz)				—	$\leq 4,0$			According to 7.5 (dielectric constant and loss tangent)		
Loss tangent (1 MHz)				—	$\leq 0,01$					
Peel strength of copper foil (Thickness of polyimide film: 25 μm)	Thickness of copper foil			—	$8 \leq$	< 18	$18 \leq$	< 35	$35 \leq$	According to 8.1 (peel strength of copper foil) (when the thickness of copper foil is less than 8 μm , electroplate copper on the foil as to attain the thickness of more than 8 μm)
	Item	Normal		N/mm	$0,6 \leq$		$0,7 \leq$		$0,8 \leq$	
		After heating								
		Dipped into solder								
		Dipped into chemicals								
Bending (Bending radius is $R = 0,38 \text{ mm}$ for thickness of polyimide film: 25 μm)	Copper foil	Type E	Copper foil	Cycle	$200 \leq$					According to 8.2 (bending)
					$100 \leq$					
		Type R			$200 \leq$					
					$100 \leq$					

Table 17 – Properties of CCL (continued)

Item			Unit	Specification		Test method JIS C 6471
Flammability			—	Specimen shall satisfy the following specification.		According to 9.1 (flammability)
				Item	Specification	
				(1) Flaming time	Within 10 s for each time, and a total of less than 50 s for 10 trials	
				(2) Flaming and glowing times	Less than 30 s for the total of second trials	
				(3) Flaming or glowing to the clamp, or marking line	There shall be no flaming or glowing	
				(4)Drop from specimen that ignite cotton	There shall be no ignition	
				NOTE Repeat the test when one specimen does not satisfy the test items (1) to (4), and the total flaming time of ten trials is between 51 s and 55 s. The specimens shall satisfy the specification in the repeated test		
Resistance to chemicals			—	There shall be no swell nor delamination		According to 9.2 (resistance to chemicals)
Resistance to soldering			—	There shall be no considerable shrinkage, no delamination of copper foil and base nor swell		According to 9.3 (resistance to soldering heat)
Solderability			—	The area of copper foil with good solderability shall be over 95 % of the foil		According to 9.4 (solderability)
Chemicals for processing of copper foil			—	The chemical for copper-foil treatment shall not affect etching of CCL, soldering nor adhesiveness of electrolytic plating		According to 9.5 (chemicals for processing of copper foil)
Dimensional stability (thickness of polyimide film: 25 μm)	After etching and drying	MD	%	Level 1: ± 0,20 Level 2: ± 0,10 Level 3: to be agreed by user and supplier Level X: to be agreed by user and supplier		According to 9.6 (dimensional stability)
		TD				
	After etching and heating	MD		Level 1: ± 0,20 Level 2: ± 0,15 Level 3: ± 0,10 Level X: ± 0075 (only for single-sided board)		
		TD				
	After heating	MD		Level 1: ± 0,20 Level 2: ± 0,10 Level 3: to be agreed by user and supplier Level X: to be agreed by user and supplier		According to Method D in Annex B of this standard
		TD				
NOTE The levels of dimensional stability are defined as: Level 1: 300 μm pitch for single-sided board and 400 μm pitch for double-sided board Level 2: 150 μm to 300 μm pitch for single-sided board and 250 μm to 400 μm pitch for double-sided board Level 3: 60 μm to 150 μm pitch for single-sided board and 125 μm to 250 μm pitch for double-sided board Level X: Less than 60 μm pitch for single-sided board						

**Table 18 – Properties of CCL –
Non-adhesive type (two layers)/polyimide film base + copper foil (sputter/plating)**

Item			Unit	Specification		Test method JIS C 6471
Observation			—	According to Table 9		According to 6.1 (observation)
Size	Thickness		—	According to 5.4.1		According to 6.2.1 (thickness)
	Width			According to 5.4.2 and 5.4.3		According to 6.2.2 (width)
	Length					According to 6.2.3 (length)
Resistivity	Normal		$\Omega \cdot \text{cm}$	$10^{13} \leq$		According to 7.1 (resistivity)
	After water vapour adsorption			$10^{12} \leq$		
Surface resistivity	Normal		Ω	$10^{11} \leq$		According to 7.2 (surface resistivity)
	After water vapour adsorption			$10^{10} \leq$		
Surface withstanding voltage AC 500 V, 1 min			—	There shall be no flash-over		According to 7.3 (surface withstanding voltage)
Interlayer withstanding voltage AC 500 V, 1 min			—	There shall be no breakdown		According to 7.4 (interlayer withstanding voltage)
Dielectric constant (1 MHz)			—	$\leq 4,0$		According to 7.5 (dielectric constant and loss tangent)
Loss tangent (1 MHz)			—	$\leq 0,07$		
Peel strength of copper foil (Thickness of polyimide film: 25 μm)	Thickness of copper foil		—	$8 \leq$	< 18	According to 8.1 (peel strength of copper foil) (when the thickness of copper foil is less than 8 μm , electroplate copper on the foil as to attain the thickness of more than 8 μm)
	Item	Normal	N/mm	$0,35 \leq$	$0,50 \leq$	
		After heating		$0,35 \leq$	$0,50 \leq$	
		Dipped into solder		$0,35 \leq$	$0,50 \leq$	
		Dipped into chemicals		To be agreed by user and supplier	$0,50 \leq$	
Bending (Bending radius is $R = 0,38\text{mm}$ for thickness of polyimide film: 25 μm)	Copper foil (plating thickness)	< 8	Cycle	To be agreed by user and supplier		According to 8.2 (bending)
		$8 \leq$ < 18		$50 \leq$		

Table 18 – Properties of CCL (continued)

Item			Unit	Specification	Test method JIS C 6471	
Flammability			—	Specimen shall satisfy the following specification	According to 9.1 (flammability)	
				Item		Specification
				(1) Flaming time		Within 10 s for each time, and a total of less than 50 s for 10 trials
				(2) Flaming and glowing times		Less than 30 s for the total of second trials
				(3) Flaming or glowing to the clamp, or marking line		There shall be no flaming or glowing
				(4) Drop from specimen that ignite cotton.		There shall be no ignition
				NOTE Repeat the test when one specimen does not satisfy the test items (1) to (4), and the total flaming time of ten trials is between 51 s and 55 s. The specimens shall satisfy the specification in the repeated test		
Resistance to chemicals			—	There shall be no swell or delamination	According to 9.2 (resistance to chemicals)	
Resistance to soldering			—	There shall be no considerable shrinkage, no delamination of copper foil and base nor swell.	According to 9.3 (resistance to soldering heat)	
Solderability			—	The area of copper foil with good solderability shall be over 95 % of the foil. (When the foil thickness is less than 8 μm, the copper shall be plated to the total thickness of more than 8 μm.)	According to 9.4 (solderability)	
Chemicals for processing of copper foil			—	The chemical for copper foil treatment shall not affect etching of CCL, soldering nor adhesiveness of electrolytic plating	According to 9.5 (chemicals for processing of copper foil)	
Dimensional stability (Thickness of polyimide film: 25 μm)	After etching and drying	MD	%	Level 1: ± 0,20 Level 2: ± 0,10 Level 3: to be agreed by user and supplier Level X: ± 0075 (only for single-sided board)	According to 9.6 (Dimensional stability)	
		TD				
	After etching and heating	MD		Level 1: ± 0,20 Level 2: ± 0,15 Level 3: ± 0,10 Level X: ± 0075 (only for single-sided board)		
		TD				
	After heating	MD		Level 1: ± 0,20 Level 2: ± 0,10 Level 3: to be agreed by user and supplier Level X: to be agreed by user and supplier	According to Method D in Annex B of this standard	
		TD				
NOTE The levels of dimensional stability are defined as:						
Level 1: 300 μm pitch for single-sided board and 400 μm pitch for double-sided board						
Level 2: 150 μm to 300 μm pitch for single-sided board and 250 μm to 400 μm pitch for double-sided board						
Level 3: 60 μm to 150 μm pitch for single-sided board and 125 μm to 250 μm pitch for double-sided board						
Level X: Less than 60 μm pitch for single-sided board						

**Table 19 – Properties of CCL –
Non-adhesive type (two layers)/polyimide film base + copper foil (laminate)**

Item					Unit	Specification			Test method JIS C 6471		
Observation					—	To be agreed by user and supplier			According to 6.1 (observation)		
Size	Thickness				—	According to 5.4.1			According to 6.2.1 (thickness)		
	Width					According to 5.4.2 and 5.4.3			According to 6.2.2 (width)		
	Length								According to 6.2.3 (length)		
Resistivity	Normal				$\Omega \cdot \text{cm}$	$10^{12} \leq$			According to 7.1 (resistivity)		
	After water vapour adsorption					$10^{11} \leq$					
Surface resistivity	Normal				Ω	$10^{11} \leq$			According to 7.2 (surface resistivity)		
	After water vapour adsorption					$10^{10} \leq$					
Surface withstanding voltage AC 500 V, 1 min					—	There shall be no flash-over			According to 7.3 (surface withstanding voltage)		
Interlayer withstanding voltage AC 500 V, 1 min					—	There shall be no breakdown			According to 7.4 (interlayer withstanding voltage)		
Dielectric constant (1 MHz)					—	$\leq 4,0$			According to 7.5 (dielectric constant and loss tangent)		
Loss tangent (1 MHz)					—	$\leq 0,01$					
Peel strength of copper foil (Thickness of polyimide film: 25 μm)	Thickness of copper foil				—	8 ≤	< 18	18 ≤	< 35	35 ≤	According to 8.1 (peel strength of copper foil) (when the thickness of copper foil is less than 8 μm, electroplate copper on the foil as to attain the thickness of more than 8 μm)
	Item	Normal			N/mm	0,6 ≤		0,7 ≤		0,8 ≤	
		After heating									
		Dipped into solder									
		Dipped into chemicals									
Bending (Bending radius is R = 0,38 mm for thickness of polyimide film: 25 μm)	Copper foil	Type E	Copper foil	18	Cycle	75 ≤				According to 8.2 (bending)	
				35		To be agreed by user and supplier					
		Type R		18		150 ≤					
				35		To be agreed by user and supplier					

Table 19 – Properties of CCL (continued)

Item			Unit	Specification		Test method JIS C 6471
Flammability			—	Specimen shall satisfy the following specification		According to 9.1 (flammability)
				Item	Specification	
				(1) Flaming time	Within 10 s for each time, and a total of less than 50 s for 10 trials	
				(2) Flaming and glowing times	Less than 30 s for the total of second trials	
				(3) Flaming or glowing to the clamp, or marking line	There shall be no flaming or glowing	
				(4) Drop from specimen that ignite cotton	There shall be no ignition	
				NOTE Repeat the test when one specimen does not satisfy the test items (1) to (4), and the total flaming time of ten trials is between 51 s and 55 s. The specimens shall satisfy the specification in the repeated test		
Resistance to chemicals			—	There shall be no swell nor delamination		According to 9.2 (resistance to chemicals)
Resistance to soldering			—	There shall be no considerable shrinkage, no delamination of copper foil and base nor swell		According to 9.3 (resistance to soldering heat)
Solderability			—	The area of copper foil with good solderability shall be over 95 % of the foil		According to 9.4 (solderability)
Chemicals for processing of copper foil			—	The chemical for copper foil treatment shall not affect the etching of CCL, soldering nor adhesiveness of electrolytic plating		According to 9.5 (chemicals for processing of copper foil)
Dimensional stability (Thickness of polyimide film: 25 μm)	After etching and drying	MD	%	Level 1: ± 0,20 Level 2: ± 0,10 Level 3: to be agreed by user and supplier Level X: to be agreed by user and supplier		According to 9.6 (dimensional stability)
		TD				
	After etching and heating	MD		Level 1: ± 0,20 Level 2: ± 0,15 Level 3: ± 0,10 Level X: ± 0075 (only for single-sided board)		
		TD				
	After heating	MD		Level 1: ± 0,20 Level 2: ± 0,10 Level 3: to be agreed by user and supplier Level X: to be agreed by user and supplier		According to Method D in Annex B of this standard
		TD				
NOTE The levels of dimensional stability are defined as:						
Level 1: 300 μm pitch for single-sided board and 400 μm pitch for double-sided board						
Level 2: 150 μm to 300 μm pitch for single-sided board and 250 μm to 400 μm pitch for double-sided board						
Level 3: 60 μm to 150 μm pitch for single-sided board and 125 μm to 250 μm pitch for double-sided board						
Level X: Less than 60 μm pitch for single-sided board						

8 Package and labelling

CCL shall be packed in such a way as not to be damaged during shipping or storage. The label on the package shall be clearly legible, shall not fade easily, and shall show the following items. The label shall be attached to each product if there is a possibility of confusion of the products when the label is on the package only (machining direction, different composition of copper foil, etc.).

- a) Type designation
- b) Film direction (for example, machine direction) by an arrow
- c) Width and length
- d) Quantity
- e) Producer or its abbreviation
- f) Date of production or its abbreviation (if the date is apparent from its lot number, the date may be omitted)
- g) Production lot number

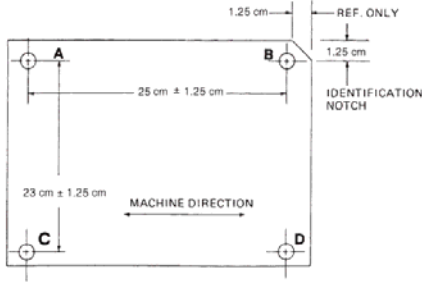
Annex A (normative)

Roughness test

Item	Roughness of shining surface of copper foil	
Specimen	Specimens of the size of 100 mm × 100 mm shall be cut from the product where abnormality is not observed	
Equipment	The test equipment specified by JIS B 0651 shall be used	
Pre-treatment	—	
Test	The specimen shall be held on the holder so that the roughness in the direction of TD can be measured and three positions for each specimen shall be measured with the conditions given below. An average value of the measurement shall be obtained.	
	*Cut-off	0,8mm
	*Measurement length	More than three times the cut-off value
	*Radius of curvature of probe	2 μm
	*Load	7 mN

Annex B (normative)

Dimensional stability test

Item	Dimensional stability (see IPC-TM-650, 2.2.4)	
Specimen	A specimen with dimensions of 27 cm × 29 cm or larger shall be cut and holes drilled as specified below (see Figure B.1)  Figure B.1 – Test pattern for the dimensional stability test	
Equipment	* Cutter: Should be able to cut a specimen larger than 27 cm × 29cm * Length measuring instrument: Should be able to read down to 0,0125 mm or an equivalent optical instrument * Drilling machine: Diameter larger than 0,889 mm * Dryer: 150 °C ± 2 °C, convection type * Etching equipment: 43 °C ± 5 °C * Marking: Maximum width of 0,125 mm * Etchant: Should not attack adhesive or insulator film	

Item	Dimensional stability (see IPC-TM-650, 2.2.4)	
Test	The test of dimensional stability after heating as specified in this standard shall be made by Method D as described below ----- Specimen: Mark lines of the width of 0,125 mm or drill holes at positions A to D (see Figure 1). Specimens are left at 23 °C ± 5 °C and 50 % ± 5 % RH for 24 h. Measure the distance between the holes or lines. Record the shortest distances between hole edges, A-B, C-D, A-C, B-D as initial distance (F₀). The time to leave specimens at ambient may be reduced if the time to reach the stable condition is known Method A: Dimensional stability of unclad material after drying by hot air (standard condition). (1) Leave specimens in a dryer for 30 min ± 2 min at 150 °C ± 2 °C. (2) Then leave specimens for 3 h at 23 °C ± 5 °C and 50 % ± 5 % RH. (3) Measure the distances between holes or lines and record the values as (F₁) after drying. Method B: Dimensional stability after etching of clad materials (1) Etch specimens, clean and dry them by air. Specimens should be in normal condition during test. (2) Then leave the specimens for 1 h at 23 °C ± 5 °C and 50 % ± 5 % RH. (3) Measure the distances between holes or lines and record the values as (F₂) after etching. (4) The dimensional stability after etching is calculated from the following equation using (F₀) and (F₂). $M.D. = \frac{\frac{(A-B)_{F_2} - (A-B)_{F_0}}{(A-B)_{F_0}} + \frac{(C-D)_{F_2} - (C-D)_{F_0}}{(C-D)_{F_0}}}{2} \times 100$ $T.D. = \frac{\frac{(A-C)_{F_2} - (A-C)_{F_0}}{(A-C)_{F_0}} + \frac{(B-D)_{F_2} - (B-D)_{F_0}}{(B-D)_{F_0}}}{2} \times 100$	

Item	Dimensional stability (see IPC-TM-650, 2.2.4)	
	Method C: Dimensional stability after etching and hot-air drying. (1) Leave specimens in a dryer for 30 min ± 2 min at 150 °C ± 2 °C after measuring them by Method B. (2) Then leave specimens for 24 h at 23 °C ± 5 °C and 50 % ± 5 % RH. (3) Measure the distances between holes or lines and record the values as (F₃) after etching and drying. (4) The dimensional stability after etching and hot air drying is calculated from the following equation using (F₀) and (F₃). $M.D. = \frac{\frac{(A-B)_{F_3} - (A-B)_{F_0}}{(A-B)_{F_0}} + \frac{(C-D)_{F_3} - (C-D)_{F_0}}{(C-D)_{F_0}}}{2} \times 100$ $T.D. = \frac{\frac{(A-C)_{F_3} - (A-C)_{F_0}}{(A-C)_{F_0}} + \frac{(B-D)_{F_3} - (B-D)_{F_0}}{(B-D)_{F_0}}}{2} \times 100$ Method D: Dimensional stability of clad materials after hot air drying (being proposed) Dimensional stability of clad materials after hot air drying is calculated from (F₂) and (F₃) as follows $M.D. = \frac{\frac{(A-B)_{F_3} - (A-B)_{F_2}}{(A-B)_{F_2}} + \frac{(C-D)_{F_3} - (C-D)_{F_2}}{(C-D)_{F_2}}}{2} \times 100$ $T.D. = \frac{\frac{(A-C)_{F_3} - (A-C)_{F_2}}{(A-C)_{F_2}} + \frac{(B-D)_{F_3} - (B-D)_{F_2}}{(B-D)_{F_2}}}{2} \times 100$ M.D.: Variation in the direction of MD T.D.: Variation in the direction of TD I: Initial value F: Measured values (F₀, F₁, F₂ and F₃) A-B: Distance between A to B A-C: Distance between A to C C-D: Distance between C to D B-D: Distance between B to D	
NOTE	* Methods A to C are stated in IPC-TC-650, 2.2.24. Method D and the equation for calculation in method D are developed by JPCA to clarify the effect of heating only to dimensional stability. * Method D is added to method A to C. It is recommended to consult the original document for methods A to C in IPC-TM-650, 2.2.4. The original IPC-TM-650 may be downloaded from the IPC home page (www.ipc.org).	

Appendix

Additional information

This information has been prepared to assist in understanding the statements made in this standard but does not form part of the standard.

I. Process of developing JPCA-BM03/CFIA-0006-2003 and revision of the first edition

Flexible printed wiring boards (hereafter called flex boards) are light-weighted and soft and are expanding their application fields for smaller and lighter electronic products. The volume of production of flex boards is nearly 20 % of the total production of printed wiring boards. It is expected that the increase of the production of flex boards will continue for years to come.

The Japan Electronic Packaging and Circuits Association organized the Standardization Committee for Flexible Wiring Boards in 1983 and developed numerous standards and specifications to provide the industry with the information required for flex boards including CCL for flex boards, a test method for cover-lays, assembly guide of electronic components on flex boards, and others.

The CCL standard of for flex boards was first published as JPCA-BM01 in 1988 as “CCL for Flexible Wiring Boards (Polyester film and polyimide film)”. This document was converted to a JIS as JIS C 6472: Copper-clad laminates for flexible printed wiring boards (polyester film, polyimide film) – (NEQ¹ IEC 60249-2-8: 1987, 60249-2-13: 1987, 60249-2-15: 1987) and widely used in the industry.

The use of flex boards in the industry is rapidly growing these days and the specifications have been extensively advanced. In many cases, conventional “three-layer structure with an adhesive layer” has been widely used in flex boards. The introduction of “two-layer type” CCL which does not use the conventional adhesive layer is a significant technical achievement in flex-board technology. The two-layer type does not use adhesive layer and has superior heat resistance to processing and dimensional stability. We did not have, however, clear standards for the two-layer type CCL, since it was new to the industry. Requests from industry for a standard for two-layer CCL have been appreciable in recent years.

The Standard Committee for Flexible Wiring Boards started collecting documentation for a revised standard for flex boards together with the support of committee members from producers of the two-layer type CCL in Japan. This document includes specifications for copper foils. We have had full support for the preparation of sections on copper foils. We agreed to publish the document as a joint standard of JPCA and of Copper Foils Industries Association (CFIA)

The present document, JPCA-BM03/CFIA006, has the nature of a revised edition of JPCA-BM01-1998. We therefore decided to disband the JPCA-BM01.

The first edition of this document was again reviewed in the Standard Committee for Flexible Wiring Boards to reflect the state of the art of the present-day flex boards together with the help of the Technical Committee of CFIA and the present JPCA-BM03/CFIA-0006-2005 was completed

The JIS document, “JIS C 6471: JIS C 6472 (1995): Copper-clad laminates for flexible printed wiring boards (polyester film, polyimide film)” should also be revised. The basic principle of the Japanese Government is to harmonize Japan Industrial Standards to IEC and ISO standards by translating the documents and adopts them as Japanese standards as a relaxation of Japanese regulations in the industry, as adopted in 1995 by the Ministry of International Trade and Industry, the Government organization at that time. We should eventually revise JIS C 6471 but the corresponding IEC document was published in 1987 and does not reflect the latest developments in the industry. IEC/TC91, the technical committee responsible for this document is in the process of revising the document. We need to maintain the JIS document until the revision of corresponding IEC standards become available.

¹ Nearly equal.

II Additional statements for some clauses of this document (clause numbers correspond to the numbers of main text)

5.2.1 Thickness and its allowance

There are two types of nominal thickness of copper foils for foil thickness of less than 5 μm ; one type for nominal thickness before surface roughing and another type for nominal thickness after surface roughing. It is therefore stated in this document that the weight-thickness of copper foil is to be agreed between the user and the supplier.

6.2 Copper foil (Tables 14 and 15)

Measurement of tensile strength and elongation for very thin copper foil is very difficult and we do not have reliable test methods. It is desired to develop appropriate test methods for these properties for thin copper film of less than 9 μm .

6.3 CCL (Tables 16 to 19)

The radius of curvature for the bending test for two-layer type CCL (Tables 17 to 19) is selected to a radius of curvature $R = 0,38 \text{ mm}$ as the two-layer CCL is very flexible. The radius for the three-layer CCL is specified as $R = 2,0 \text{ mm}$ in Table 16. A unified test condition such as $R = 0,8 \text{ mm}$ may be studied in future to develop common test conditions.

ISBN 2-8318-9161-2



ICS 31.180
